

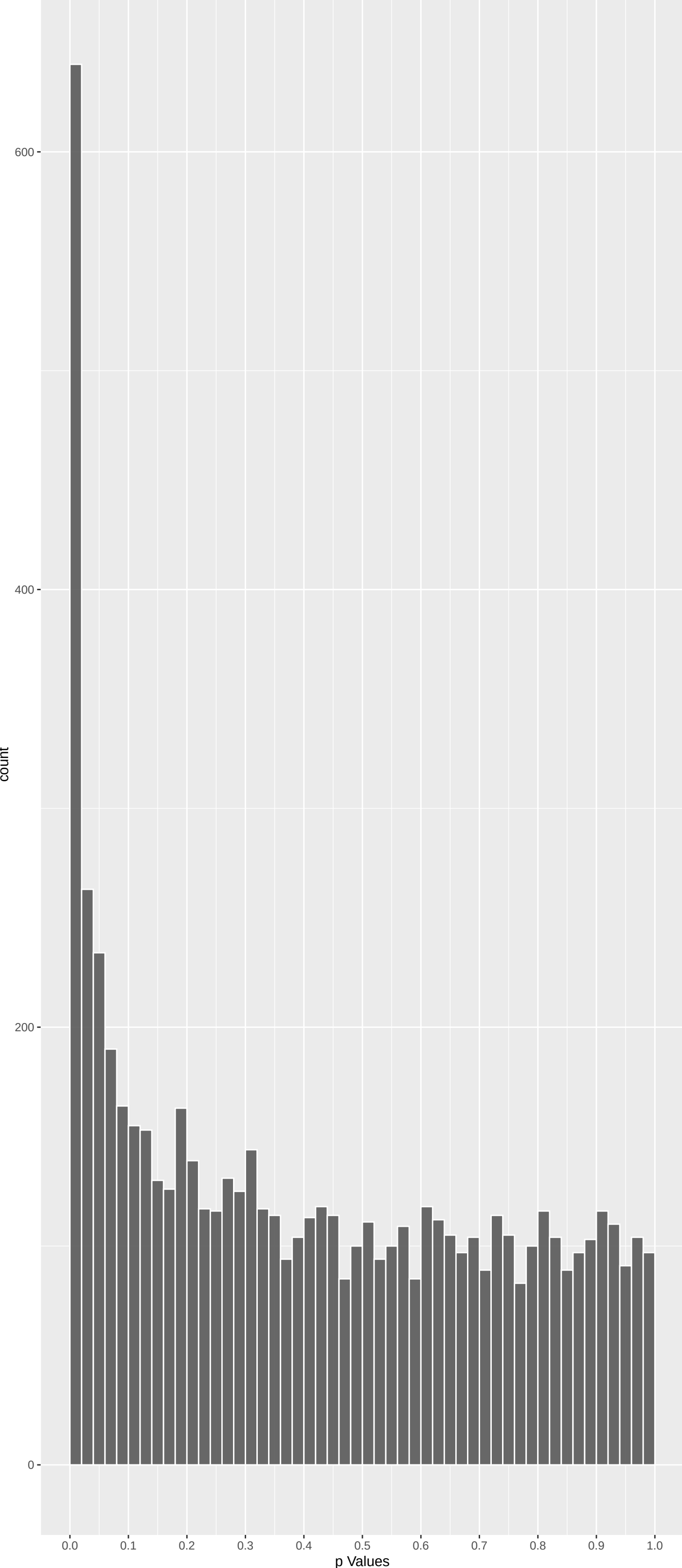
Top genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
CLDN16	-7.516993	5.605021e-14	8.470e-10	1.280e-05
BP1FB1	-7.005972	2.452764e-12	1.853e-08	1.400e-04
SLC5A12	-6.705583	2.006037e-11	1.011e-07	5.090e-04
CPB1	-6.436361	1.223720e-10	4.623e-07	1.747e-03
EHHADH	-6.055063	1.403625e-09	3.784e-06	9.283e-03
SLC23A1	-6.044102	1.502448e-09	3.784e-06	9.283e-03
HMCES	-5.998476	1.991778e-09	4.300e-06	9.283e-03
PNLIP	-5.877231	4.171858e-09	7.881e-06	1.100e-02
GCDH	-5.821671	5.826226e-09	8.004e-06	1.100e-02
CRYAA	5.836265	5.338387e-09	8.004e-06	1.100e-02
ALB	-5.841558	5.171482e-09	8.004e-06	1.100e-02
PPP4R1	-5.725148	1.033439e-08	1.301e-05	1.639e-02
CLPS	-5.683709	1.318042e-08	1.532e-05	1.781e-02
CTRC	-5.651030	1.594896e-08	1.722e-05	1.858e-02
SLC14A2	-5.592943	2.232531e-08	2.249e-05	2.266e-02
BNC1	5.545886	2.924693e-08	2.762e-05	2.609e-02
H4-16	5.467480	4.564783e-08	4.058e-05	3.607e-02
MGST3	-5.428776	5.674192e-08	4.764e-05	3.740e-02
ANK1	-5.416207	6.087669e-08	4.842e-05	3.740e-02
SLC4A1	-5.386144	7.198514e-08	5.198e-05	3.740e-02
CAMK2B	5.385535	7.222941e-08	5.198e-05	3.740e-02
PGC	-5.335439	9.531345e-08	6.547e-05	4.497e-02
PGA4	-5.297668	1.172911e-07	7.707e-05	5.064e-02
TAS2R10	-5.273580	1.337880e-07	8.424e-05	5.304e-02
CPLX3	5.239893	1.606698e-07	9.712e-05	5.813e-02

Top genes by Q-Value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
CLDN16	-7.516993	5.605021e-14	8.470e-10	1.280e-05
BP1FB1	-7.005972	2.452764e-12	1.853e-08	1.400e-04
SLC5A12	-6.705583	2.006037e-11	1.011e-07	5.090e-04
CPB1	-6.436361	1.223720e-10	4.623e-07	1.747e-03
EHHADH	-6.055063	1.403625e-09	3.784e-06	9.283e-03
SLC23A1	-6.044102	1.502448e-09	3.784e-06	9.283e-03
HMCES	-5.998476	1.991778e-09	4.300e-06	9.283e-03
GCDH	-5.821671	5.826226e-09	8.004e-06	1.100e-02
CRYAA	5.836265	5.338387e-09	8.004e-06	1.100e-02
ALB	-5.841558	5.171482e-09	8.004e-06	1.100e-02
PNLIP	-5.877231	4.171858e-09	7.881e-06	1.100e-02
PPP4R1	-5.725148	1.033439e-08	1.301e-05	1.639e-02
CLPS	-5.683709	1.318042e-08	1.532e-05	1.781e-02
CTRC	-5.651030	1.594896e-08	1.722e-05	1.858e-02
SLC14A2	-5.592943	2.232531e-08	2.249e-05	2.266e-02
BNC1	5.545886	2.924693e-08	2.762e-05	2.609e-02
H4-16	5.467480	4.564783e-08	4.058e-05	3.607e-02
MGST3	-5.386144	7.198514e-08	5.198e-05	3.740e-02
ANK1	-5.416207	6.087669e-08	4.842e-05	3.740e-02
SLC4A1	-5.428776	5.674192e-08	4.764e-05	3.740e-02
CAMK2B	5.385535	7.222941e-08	5.198e-05	3.740e-02
PGC	-5.335439	9.531345e-08	6.547e-05	4.497e-02
PGA4	-5.297668	1.172911e-07	7.707e-05	5.064e-02
TAS2R10	-5.273580	1.337880e-07	8.424e-05	5.304e-02
SLC10A2	-5.091377	3.554720e-07	1.462e-04	5.813e-02

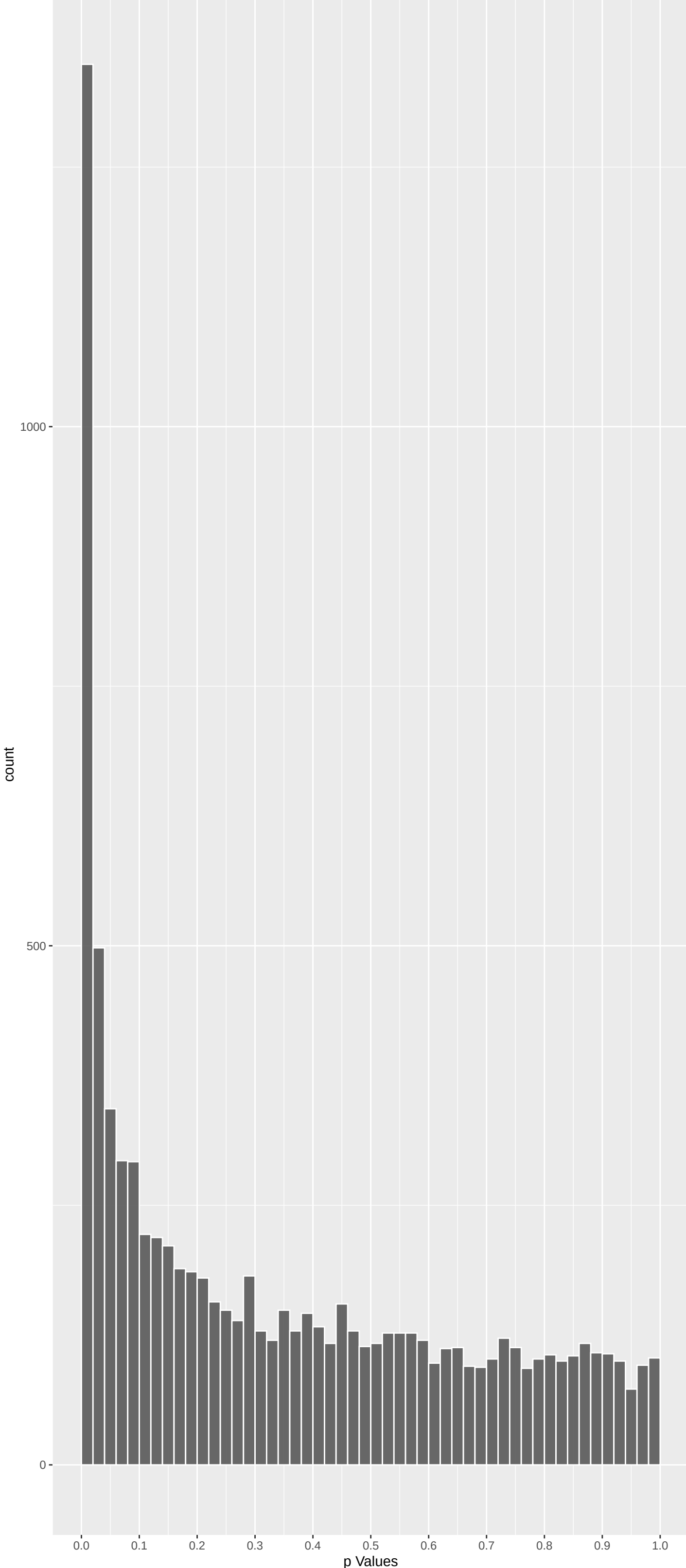
Positive Rho Non-permulated



Top Positive genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
CRYAA	5.836265	5.338387e-09	8.004e-06	1.100e-02
BNC1	5.545886	2.924693e-08	2.762e-05	2.609e-02
H4-16	5.467480	4.564783e-08	4.058e-05	3.607e-02
CAMK2B	5.385535	7.222941e-08	5.198e-05	3.740e-02
CPLX3	5.239893	1.606698e-07	9.712e-05	5.813e-02
CFAP20	5.145717	2.665008e-07	1.305e-04	5.813e-02
SRSF3	4.958967	7.086916e-07	2.136e-04	6.329e-02
CRYAA2	4.955662	7.208437e-07	2.136e-04	6.329e-02
RPL34	4.892749	9.943723e-07	2.683e-04	7.004e-02
SNRPD3	4.817692	1.452286e-06	3.325e-04	7.614e-02

Negative Rho Non-permulated



Top Negative genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
CLDN16	-7.516993	5.605021e-14	8.470e-10	1.280e-05
BP1FB1	-7.005972	2.452764e-12	1.853e-08	1.400e-04
SLC5A12	-6.705583	2.006037e-11	1.011e-07	5.090e-04
CPB1	-6.436361	1.223720e-10	4.623e-07	1.747e-03
EHHADH	-6.055063	1.403625e-09	3.784e-06	9.283e-03
SLC23A1	-6.044102	1.502448e-09	3.784e-06	9.283e-03
HMCES	-5.998476	1.991778e-09	4.300e-06	9.283e-03
PNLIP	-5.877231	4.171858e-09	7.881e-06	1.100e-02
GCDH	-5.821671	5.826226e-09	8.004e-06	1.100e-02
ALB	-5.841558	5.171482e-09	8.004e-06	1.100e-02

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Fatty Acid Metabolic Process (GO:0006631)	-0.21034135	108	4.808e-14	2.590e-10	CYP1A1:41 ABCB11:82 CPT1A:139 PLA2G1B:162 ABCD3:276 ACADSB:297
Fatty Acid Catabolic Process (GO:0009062)	-0.26234838	58	4.529e-12	1.220e-08	EHHADH:5 EC12:151 ABCD3:276 ACADH:312 ACAD11:371 DEC1:380
Fatty Acid Beta-Oxidation (GO:0006635)	-0.27437488	47	7.889e-11	1.417e-07	EHHADH:5 GCDH:10 EC12:151 ABCD3:276 ACADM:312 ACAD11:371
Monocarboxylic Acid Transport (GO:001571)	-0.22901951	61	6.416e-10	8.641e-07	SLC5A12:3 SLC16A5:25 SLC10A2:30 SLC22A13:32 SLC6A12:36 FABP1:72
Organic Hydroxy Compound Transport (GO:0)	-0.29637363	35	1.323e-09	1.426e-06	SLC10A2:30 SLC51A:215 ATP8B1:229 ABCD3:276 RALBP1:286 ABCG5:296
Fatty Acid Oxidation (GO:0019395)	-0.25441988	47	1.642e-09	1.474e-06	EHHADH:5 EC12:151 ABCD3:276 ACADM:312 ACAD11:371 DEC1:380
Monocarboxylic Acid Metabolic Process (G	-0.18461497	80	1.196e-08	9.203e-06	AGXT:39 CYP1A1:41 ABCB11:82 HOGA1:103 NPC1:125 CPT1A:139
Nitrogen Compound Transport (GO:0071705)	-0.13734075	144	1.397e-08	9.406e-06	SLC22A13:32 SLC6A12:36 SLC6A6:40 RHAG:53 RHD:129 RHCE:143
Alpha-Amino Acid Catabolic Process (GO:1)	-0.29766639	29	2.926e-08	1.588e-05	TAT:24 GOT2:93 HOGA1:103 MCCCL1:204 SARHD:213 HIBCH:223
Steroid Metabolic Process (GO:0008202)	-0.20547412	61	2.948e-08	1.588e-05	CYP11A1:22 DHRS4:33 CYP1A1:41 CYP17A1:63 NPC1:125 NR12:158
Chemical Synaptic Transmission (GO:00072	0.1009280	237	9.769e-08	4.784e-05	SNAP25:11 GABRR1:56 NPTX1:57 GRK13:102 SYT1:113 GRK1:165
Modulation Of Chemical Synaptic Transmis	0.14917660	105	1.355e-07	6.082e-05	CLX3:5 NPTX1:57 GRK13:102 SYT1:113 LRRCA:163 GRK1:165
Amino Acid Catabolic Process (GO:0009063	-0.23582970	38	4.967e-07	2.058e-04	GOT2:93 HOGA1:103 MCCCL1:204 SARHD:213 HIBCH:223 BCKDK:767
Cell Morphogenesis Involved In Neuron Di	0.18118493	64	5.522e-07	2.125e-04	SLITRK4:70 SLITRK1:138 TRIM46:166 NRCAM:173 CNTN4:359 POU4F1:636
Anterograde Trans-Synaptic Signaling (GO	0.11126573	169	6.550e-07	2.352e-04	SNAP25:11 GABRR1:56 NPTX1:57 SYT1:113 GRK1:165 CHRM2:196
Regulation Of Heart Rate By Cardiac Cond	0.23265996	36	1.378e-06	4.641e-04	KCNDB3:69 KCNH6:241 SCN3B:398 CAV1:644 BIN1:1043 KCNE3:1058
Nervous System Development (GO:0007399)	0.0740031	363	1.501e-06	4.756e-04	MAB21L2:12 DCLK1:55 NPTX1:57 MOG:71 VAX2:98 SLITRK1:138
Cardiac Conduction (GO:0061337)	0.21623555	41	1.687e-06	5.048e-04	KCNDB3:69 KCNH6:241 SCN3B:398 CAV1:644 BIN1:1043 KCNE3:1058
Bile Acid And Bile Salt Transport (GO:00	-0.34025941	16	2.461e-06	6.977e-04	SLC10A2:30 SLC51A:215 ATP8B1:229 ABCD3:276 SLC01B1:300 SLC10A6:580
Neuron Projection Morphogenesis (GO:0048	0.11946541	124	4.568e-06	1.158e-03	EPHA4:41 DCLK1:55 SLITRK4:70 CDC42:77 SLITRK1:138 TRIM46:166
Sensory Perception Of Light Stimulus (GO	0.13930157	91	4.535e-06	1.158e-03	CRYAA:1 CRYBA1:36 CRYBB2:43 GNAT2:78 VAX2:98 TRPM1:111
Visual Perception (GO:0007601)	0.13980297	90	4.727e-06	1.158e-03	CRYAA:1 CRYBA1:36 CRYBB2:43 GNAT2:78 VAX2:98 TRPM1:111
Phosphatidylcholine Metabolic Process (G	-0.19321026	46	5.890e-06	1.379e-03	PNLIPRP2:46 PLA2G1B:162 ABHD3:188 SLC44A4:222 PLA2G2D:278 TSP0:757
Alpha-Amino Acid Biosynthesis Process (G	-0.4118960	10	6.489e-06	1.455e-03	SDS:47 BCAT2:114 CPS1:261 SDSL:383 ILVB1:1117 BCAT1:1461
Regulation Of Synaptic Transmission, Glu	0.18992748	47	6.754e-06	1.455e-03	GRK13:102 SYT1:113 GRK1:165 GRMD:2199 GRM8:239 GRIN2B:294
Regulation Of Neuron Projection Developm	0.10403049	156	7.834e-06	1.623e-03	CAMK2B:4 ALKAL1:40 EPHA4:41 NCSL1:48 NDLE1:148 ATP1B2:211
Quaternary Ammonium Group Transport (GO:	-0.136385737	12	1.278e-05	2.549e-03	SLC44A4:222 SLC22A4:263 SLC22A1:379 SLC6A20:576 SLC14A5:1051 SLC5A7:1554
Regulation Of Transcription By RNA Polym	0.03406669	1527	1.702e-05	3.274e-03	UBE2D1:15 UBE2I:19 HSF4:26 RPL7:38 CRTC3:54 ELOC:72
Xonogenesis (GO:0007409)	0.09722793	164	1.845e-05	3.314e-03	EPHA4:41 SLITRK4:70 SLITRK1:138 TRIM46:166 NRCAM:173 ULK2:301
One-Carbon Compound Transport (GO:001975	-0.22997048	29	1.831e-05	3.314e-03	SLC14A2:14 SLC4A1:17 RHAG:53 CAA1:44 CA2:146 SLC4A7:1189
Fatty Acid Transport (GO:0015908)	-0.20759219	35	2.158e-05	3.695e-03	FABP1:72 GOT2:93 SLC22A1:379 SLC22A4:263 PPARD:866 SLC22A6:912
Sensory Perception Of Smell (GO:00007608)	-0.15239075	65	2.195e-05	3.695e-03	OR13C9:167 OMP:736 OR4D1:800 OR21:847 OR6N2:925 OR5B21:1158
Axon Development (GO:0061564)	0.13349866	84	2.410e-05	3.873e-03	SLITRK4:70 SLITRK1:138 TRIM46:166 NRCAM:173 CNTN4:359 GPM6B:375
Ethanol Metabolic Process (GO:0006067)	-0.43085401	8	2.444e-05	3.873e-03	SULT1A2:132 SULT1A3:138 SULT1E1:170 SULT1B1:848 ACSG2:856 ADH4:1199
Carboxylic Acid Transport (GO:0046942)	-0.17401426	49	2.547e-05	3.920e-03	SLC5A12:3 SLC16A5:25 SLC6A12:36 SLC6A6:40 SLC6A19:169 SLC43A1:179
Glutamate Receptor Signaling Pathway (GO	0.21620676	31	3.122e-05	4.672e-03	GRK13:102 TRPM1:111 HOMER2:133 GRK1:165 GRM2:199 GRM8:239
Organic Cation Transport (GO:0015695)	-0.21388883	31	3.793e-05	5.523e-03	SLC6A12:36 SLC44A4:222 SLC22A4:263 RALBP1:286 TSP02:370 SLC22A1:379
Branched-Chain Amino Acid Metabolic Proc	-0.30101820	15	5.441e-05	7.713e-03	BCAT2:114 MCCCL1:204 HIBCH:223 RALBK:767 PCGB:1050 IVD:1094
Lipid Transport (GO:0006869)	-0.12413848	88	5.852e-05	7.881e-03	EHHADH:5 SLC10A2:30 FABP1:72 ABCB11:82 GOT2:93 MTPP:123
Regulation Of Cell Differentiation (GO:0	0.09297235	158	5.795e-05	7.881e-03	FGF8:46 PHLD1B:167 TEN15C:143 DDK17:184 MTA3:216 MTA1:518

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
HSIAO_LIVER_SPECIFIC_GENES	-0.18866666	196	1.004e-19	6.511e-16	ALB:9 TAT:24 AGXT:39 SDS:47 ALDH1A1:56 FABP1:72
CARRILLORIEACH_HEPATOBlastOMA_VS_NORMAL	-0.08449157	911	1.162e-17	2.524e-14	EHHADH:5 HMCE5:7 GCDH:10 CYP11A1:22 TAT:24 GNMT:29
REACTOME_METABOLISM_OF_LIPIDS	-0.09901782	651	1.167e-17	2.524e-14	EHHADH:5 ALB:9 CYP11A1:22 SLC10A2:30 CYP1A1:41 SPTLC3:43
REACTOME_FATTY_ACID_METABOLISM	-0.18539336	155	1.818e-15	2.948e-12	EHHADH:5 CYP1A1:41 ELVOL:7:85 SCD:92 CPT1A:139 EC12:151
REACTOME_OXIDATIVE_STRESS_INDUCED_SENESC	0.25351230	76	2.243e-14	2.909e-11	RPS27A:14 UBA52:20 HAC2:34 HAC14:45 H3C13:59 HAC15:75
REACTOME_BIOLOGICAL_OXIDATIONS	-0.17607463	146	2.248e-13	2.431e-10	MGST3:15 CYP11A1:22 CYP1A1:41 ALDH1A1:56 CYP2F1:57 CHAC2:66
REACTOME_NEURONAL_SYSTEM	0.11350397	352	3.144e-13	2.549e-10	CAMK2B:4 SNAP25:11 ADCY1:37 PRKCG:53 GABRR1:56 KCND3:69
REACTOME_OLFACTORY_SIGNALING_PATHWAY	-0.20619519	105	3.034e-13	2.549e-10	OR13C9:167 OR10V1:230 OR5AR1:245 OR52K2:421 OR4D11:731 OR4D1:800
REACTOME_DEVELOPMENTAL_BIOLOGY	0.07511183	792	1.056e-12	7.612e-10	RPL34:9 RPS27A:14 POLR2J1:17 UBA52:20 MAGOH:21 HAC2:34
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	-0.13263648	232	3.842e-12	2.492e-09	SLC5A12:3 SLC14A2:14 SLC4A1:17 SLC6A12:36 SLC6A6:40 SLC13A2:50
REACTOME_CELLULAR_SENESCENCE	0.1728444	135	5.118e-12	3.018e-09	RPS27A:14 UBE2D1:15 UBA52:20 HAC2:34 HAC14:45 H3C13:59
REACTOME_NERVOUS_SYSTEM_DEVELOPMENT	0.09596827	436	8.025e-12	4.338e-09	RPL34:9 RPS27A:14 UBA52:20 MAGOH:21 RPL1:37 EPHA4:41
KEGG_VALINE_LEUCINE_AND_Isoleucine_DEGRA	-0.29742758	43	1.518e-11	7.575e-09	EHHADH:5 ACAD8:104 BCAT2:114 MCCCL1:204 HIBCH:223 ACADSB:297
REACTOME_CELLULAR_RESPONSES_TO_STIMULI	0.08031929	589	3.687e-11	1.708e-08	CAMK2B:4 RPL34:9 RPS27A:14 UBE2D1:15 UBA52:20 HAC2:34
REACTOME_NEUROTRANSMITTER_RECEPTORS_AND_	0.15162121	159	4.491e-11	1.821e-08	CAMK2B:4 ADCY1:37 PRKCG:53 GABRR1:56 GRK13:102 NBEA:125
CAIRO_HEPATOBlastOMA_DN	-0.12593522	232	4.336e-11	1.821e-08	GCDH:10 GNMT:29 AGXT:39 SDS:47 TMPRSS2:51 SLC1A1:60
REACTOME_METABOLISM_OF_AMINO_ACIDS_AND_D	-0.11050687	299	5.689e-11	1.942e-08	GCDH:10 TAT:24 GNMT:29 SLC6A12:36 AGXT:39 SDS:47
CHIANG_LIVER_CANCER_SUBCLASS_PROLIFERATI	-0.15486546	151	5.446e-11	1.942e-08	EHHADH:5 GCDH:10 TAT:24 AGXT:39 ABCB11:82 DHTKD1:98
KEGG_OLFACTORY_TRANSDUCTION	-0.17418552	119	5.547e-11	1.942e-08	OR13C9:167 OR10V1:230 OR5AR1:245 CLCA1:377 OR52K2:421 CALM2:502
BENPORATH_ES_WITH_H3K27ME3	0.06390560	913	9.263e-11	3.005e-08	BNC1:2 CAMK2B:4 MAB21L2:12 HSF4:26 EGLFAM:35 ALKAL1:40
WP_MALIGNANT_PLEURAL_MESOTHELIOMA	0.09759329	369	1.434e-10	4.431e-08	ACTC1:28 FGF8:46 ACTA2:85 ITGB4:91 EED:109 ACTG2:130
REACTOME_INTRACELLULAR_SIGNALING_BY_SECO	0.11758207	250	1.712e-10	5.047e-08	CAMK2B:4 RPS27A:14 UBA52:20 ADCY1:37 FGF8:46 PRKCG:53
WOO_LIVER_CANCER_RECURRENCE_DN	-0.22251131	66	4.163e-10	1.174e-07	GCDH:10 ADCY1:83 GOT2:93 MTPP:123 HGD:124 SLC2A9:202
BENPORATH_SUZ12_TARGETS	0.06269669	852	7.580e-10	2.049e-07	BNC1:2 MAB21L2:12 HSF4:26 EGLFAM:35 ALKAL1:40 EPHA4:41
REACTOME_SIGNALING_BY_RECEPTOR_TYROSINE_	0.08609907	436	8.525e-10	2.212e-07	RPS27A:14 POLR2J1:17 UBA52:20 FGF8:46 RHOA:52 MAPKAP1:73
LEE_LIVER_CANCER_DENA_DN	-0.21816056	66	9.033e-10	2.254e-07	EHHADH:5 PNLIP:8 GCDH:10 CTRC1:13 ALDH1A1:56 CYP2F1:57
KIM_ALL_DISORDERS_OLIGODENDROCYTE_NUMBER	0.07355633	603	9.904e-10	2.295e-07	CAMK2B:4 UBE2M:13 RPS27A:14 POLR2J1:17 RACK1:39 RAN:44
KIM_ALL_DISORDERS_CALB1_CORR_UP	0.08522208	442	9.796e-10	2.295e-07	CAMK2B:4 SNAP25:11 VSNL1:31 RAN:44 DCLK1:55 NPTX1:57
KIM_BIPOLAR_DISORDER_OLIGODENDROCYTE_DEN	0.07726247	831	1.402e-09	3.137e-07	CAMK2B:4 RPS27A:14 POLR2J1:17 VSNL1:31 RAN:44 RHOA:52
BENPORATH_EED_TARGETS	0.06044436	560	2.546e-09	5.505e-07	BNC1:2 CAMK2B:4 MAB21L2:12 HSF4:26 TM6SF1:30 HAC2:34
LIM_MAMMARY_STEM_CELL_UP	0.08328880	433	3.324e-09	6.955e-07	BNC1:2 VSNL1:31 PHLD1B:167 ACTA2:85 PLP1:88 ITGB4:91
REACTOME_MAPK_FAMILY_SIGNALING_CASCADES	0.10476050	264	5.120e-09	1.038e-06	CAMK2B:4 RPS27A:14 UBA52:20 FGF8:46 ICMT:50 CDC42:77
HOSHIDA_LIVER_CANCER_SUBCLASS_S3	-0.11037430	235	6.092e-09	1.198e-06	EHHADH:5 SLC23A1:6 SLC6A12:36 AGXT:39 SDS:47 ALDH1A1:56
KEGG_GLYCINE_SERINE_AND_THREONINE_METABO	-0.30131013	31	6.431e-09	1.227e-06	GNMT:29 AGXT:39 SDS:47 SARHD:213 BHMAT:225 DMGDH:232
HOUNKPE_HOUSEKEEPING_GENES	0.05879803	860	6.784e-09	1.255e-06	SRSF3:7 SNRPD:10 POLR2J1:17 SF3B8:18 UBE2I:19 MAGOH:21
REACTOME_SIGNALING_BY_WNT	0.11675730	208	6.967e-09	1.255e-06	RPS27A:14 UBA52:20 HAC2:34 HAC14:45 RHOA:52 PRKCG:53
NIKOLSKY_BREAST_CANCER_8P12_P11_AMPLICON	-0.23607891	50	7.800e-09	1.358e-06	ANK1:16 ADAM2:21 IDO2:91 NSD5:118 ADAM9:119 CHRNA6:178
KEGG_FATTY_ACID_METABOLISM	-0.27792881	36	7.956e-09	1.358e-06	EHHADH:5 GCDH:10 CPT1A:139 EC12:151 ACADSB:297 ACADM:312
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	0.15168660	120	9.899e-09	1.647e-06	DNAJC5:151 ARFRP1:167 LAMA5:263 ADRM1:319 CHRNA4:339 CBLN4:358
REACTOME_ESR_MEDIATED_SIGNALING	0.14476716	131	1.103e-08	1.789e-06	POLR2J1:17 HAC2:34 HAC14:45 H3C13:59 HAC15:75 H2BC15:97

DisGenET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Hyperammonemia	-0.17859367	117	2.756e-11	2.702e-07	MANBA:55 NPC1:125 NAGLU:131 CPT1A:139 MCCCL1:204 NAGS:211
Drug-Induced Liver Disease	-0.11552404	200	1.999e-08	9.798e-05	ALB:9 CYP1A1:41 IL1R2:49 CA3:71 ABCB11:82 CRYZ:107
Cystic Fibrosis	-0.07723043	421	6.952e-08	2.272e-04	BP1FB1:2 ALB:9 CTRC:13 SDS:47 LCT:86 GOT2:93
Cholestasis	-0.09594453	247	2.348e-07	5.755e-04	SLC23A1:6 SLC10A2:30 CYP17A1:63 ABCB11:82 HOGA1:103 NPC1:125
Renal salt wasting	-0.37348283	15	5.515e-07	1.081e-03	CYP11A1:22 KCNJ1:37 CLCNKA:108 CLCNKB:134 SCNN1A:212 SCNN1B:217
Metabolic acidosis	-0.17751203	65	7.652e-07	1.250e-03	EHHADH:5 GCDH:10 SLC4A1:17 AGXT:39 LCT:86 SLC9A4:99
HYPERGLYCINURIA (disorder)	-0.36616279	15	9.139e-07	1.280e-03	SLC6A19:169 ACADM:312 SLC6A20:576 GCSH:921 PCCB:1050 ALDH4A1:1062
Kidney Failure, Chronic	-0.07624834	345	1.331e-06	1.290e-03	ALB:9 SLC4A1:17 KCNJ1:37 AGXT:39 CYP1A1:41 FABP1:72
Hepatomegaly	-0.08779913	325	1.210e-06	1.290e-03	EHHADH:5 GCDH:10 GNMT:29 CYP1A1:41 RHAG:53 ABCB11:82
Steatohepatitis	-0.06626893	458	1.447e-06	1.290e-03	ALB:9 SLC4A1:17 CA3:71 FABP1:72 SCD:92 MTP:123
Vomiting	-0.10946579	167	1.138e-06	1.290e-03	GCDH:10 CYP11A1:22 KCNJ1:37 COMT:74 XPC:101 HTR3C:141
Glycinuria	-0.38189413	13	1.869e-06	1.527e-03	SLC6A19:169 ACADM:312 SLC6A20:576 GCSH:921 PCCB:1050 ALDH4A1:1062
Fabry Disease	-0.21396582	41	2.161e-06	1.630e-03	MANBA:55 CHDH:241 SI:267 VDR:305 ABCB1:335 GLA:61
Autism Spectrum Disorders	0.06469175	456	2.681e-06	1.878e-03	SNAP25:11 IQGAP3:16 ACTC1:28 JAKMIP1:32 ETF1:47 NCS1:48
Hepatitis, Toxic	-0.10528953	161	4.291e-06	1.913e-03	ALB:9 IL1R2:49 CA3:71 NR1I2:158 SULT1E1:170 CAT:240
Chemical and Drug Induced Liver Injury	-0.10528953	161	4.291e-06	1.913e-03	ALB:9 IL1R2:49 CA3:71 NR1I2:158 SULT1E1:170 CAT:240
Chemically-Induced Liver Toxicity	-0.10528953	161	4.291e-06	1.913e-03	ALB:9 IL1R2:49 CA3:71 NR1I2:158 SULT1E1:170 CAT:240
Drug-Induced Acute Liver Injury	-0.10528953	161	4.291e-06	1.913e-03	ALB:9 IL1R2:49 CA3:71 NR1I2:158 SULT1E1:170 CAT:240
Hyperactive renin-angiotensin system	-0.47123765	8	3.923e-06	1.913e-03	KCNJ1:37 CLCNKB:134 SLC26A3:203 SCNN1A:212 SCNN1B:217 SCNN1G:772
Inborn Errors of Metabolism	-0.16452708	66	3.887e-06	1.913e-03	GCDH:10 AGXT:39 ACY1:83 HEXB:94 HGD:124 CA2:146
Increased plasma renin activity	-0.47123765	8	3.923e-06	1.913e-03	KCNJ1:37 CLCNKB:134 SLC26A3:203 SCNN1A:212 SCNN1B:217 SCNN1G:772
Lysosomal Storage Diseases	-0.15733284	72	4.015e-06	1.913e-03	MANBA:55 HEXB:94 NPC1:125 NAGLU:131 SGGC:155 RPS27:253
Pancreatitis	-0.10378062	165	4.517e-06	1.926e-03	PNLIP:9 ALB:9 CTRC:13 CYP1A1:41 PNLIPRP2:46 CELA3B:117
Kidney Calculi	-0.18184705	53	4.746e-06	1.939e-03	CLDN16:1 SLC4A1:17 AGXT:39 SLC13A2:50 SLC2A9:202 CAT:240
Hepatitis, Drug-Induced	-0.10337933	164	5.359e-06	2.102e-03	ALB:9 IL1R2:49 CA3:71 NR1I2:158 SULT1E1:170 CAT:240
Atherogenesis	-0.19821912	43	6.901e-06	2.603e-03	NPC1:125 ABCG5:296 TRPV1:309 HHRH:356 CNR2:524 PTGS2:578
Metabolic Diseases	-0.06990403	349	8.285e-06	3.09e-03	GCDH:10 TAT:24 AGXT:39 ACY1:83 SCD:92 HGD:124
Hyperlipidemia	-0.05957714	181	9.034e-06	3.163e-03	ALB:9 MTP:123 NR1I2:158 PLA2G1B:162 CYP7A1:220 CHDH:241
Hereditary spherocytosis	-0.26604779	23	1.008e-05	3.407e-03	ANK1:16 SLC4A1:17 SDS:47 SPTA1:48 EPB42:214 SLC26A8:257
Tetany	-0.28319069	20	1.169e-05	3.820e-03	CLDN16:1 KCNJ1:37 HIRA:68 COMT:74 TRPM6:78 SLC12A3:249
Coronary Artery Disease	-0.04629911	793	1.270e-05	4.017e-03	ALB:9 CYP1A1:41 PROM1:45 CYP17A1:63 PGD5:70 COMT:74
Hyperkalemia	-0.30109555	17	1.279e-05	5.298e-03	CYP11A1:22 CYP17A1:63 WNK4:197 SCNN1A:212 SCNN1B:217 CA12:666
Malabsorption Syndrome	-0.21132381	34	2.023e-05	6.011e-03	PNLIP:9 SLC10A2:30 FABP1:72 LCT:86 MTP:123 CUBN:161
Dyslipidemias	-0.09172485	181	2.211e-05	6.377e-03	ALB:9 SCD:92 GOT2:93 MTP:123 RORAL:140 NR1I2:158
Peripheral Vascular Diseases	-0.14229186	72	3.048e-05	8.540e-03	ALB:9 AGXT:39 CYP1A1:41 MTP:123 ABCG5:296 ABCG8:406
Hypercalciuria	-0.18549700	41	4.002e-05	1.090e-02	CLDN16:1 KCNJ1:37 WNK4:197 SLC12A3:249 VDR:305 PTH:337
Nonalcoholic Steatohepatitis	-0.19056562	168	4.906e-05	1.300e-02	CYP17A1:63 MTP:123 CPT1A:139 LAMP1:350 DECR1:380 SIRT6:428
Diabetic Retinopathy	-0.08224007	203	5.709e-05	1.386e-02	ALB:9 SDS:47 PGF:154 SARDH:213 CAT:240 SLC12A3:249
Hypocalcemia	-0.17779733	43	5.550e-05	1.386e-02	CLDN16:1 SLC4A1:17 HIRA:68 COMT:74 TRPM6:78 CHDH:241
Hypochloremia (disorder)	-0.47290232	6	6.032e-05	1.386e-02	KCNJ1:37 CLCNKA:108 CLCNKB:134 SLC26A3:203 SLC12A1:867 BSND:1023